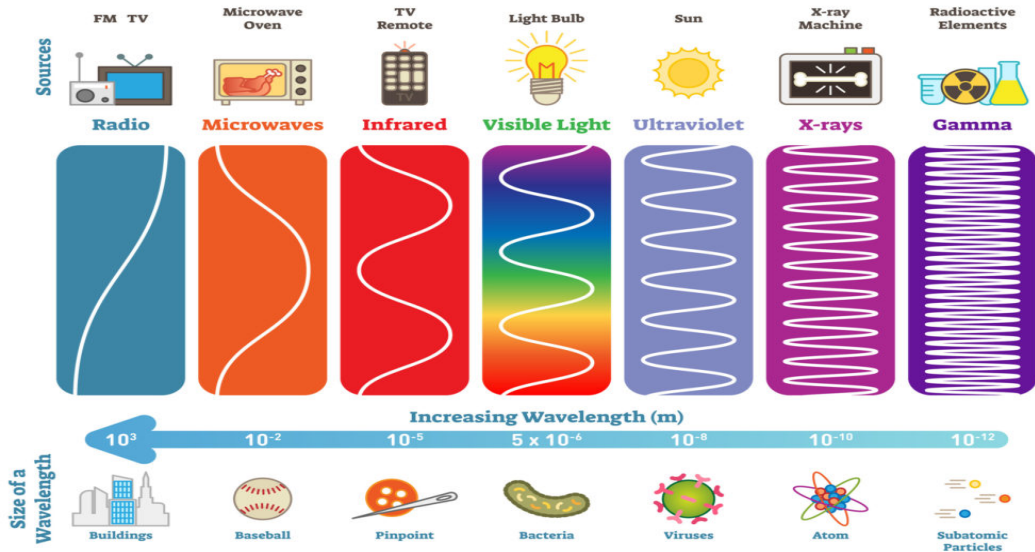




ELECTROMAGNETIC WAVES

CHAPTER EIGHT CLASS - XII

MANISH JOSHI

D.A.V. LOHAGHAT

August 24, 2025

Learning Objectives:

- Introduction
- Basic idea of displacement current,
- Electromagnetic waves, their characteristics, their transverse nature (qualitative idea only).
- Electromagnetic spectrum (radio waves, microwaves, infra-red, visible, ultraviolet, X-rays, gamma rays) including elementary facts about their uses.

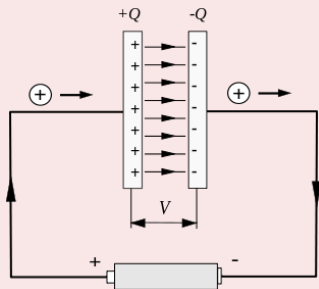
INTRODUCTION:

- The phenomenon of Faraday's electromagnetic induction concludes that a **changing magnetic field** at a point with time **produces an electric field** at that point.
- **Maxwell**, pointed out that there is a **symmetry in nature** (i.e) **changing electric field** with time at a point **produces a magnetic field** at that point.
- It means that a **change in one field with time** (either electric or magnetic) **produces another field**.
- This idea led Maxwell to conclude that the **variation in electric and magnetic fields perpendicular to each other, produces electromagnetic disturbances** in space.

INTRODUCTION:

- **These disturbances** have the properties of a wave and **propagate through space without any material medium.**
- These waves are called **electromagnetic waves**. They travel in vacuum or free space with a velocity $3 \times 10^8 \text{ ms}^{-1}$

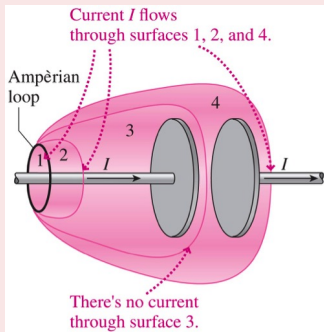
DISPLACEMENT CURRENT:



- We know that an electric current produces a magnetic field around it. **Maxwell showed that a changing electric field must also produce a magnetic field .**
- To see how a changing electric field gives rise to a magnetic field, let us consider the process of **charging of a capacitor.**
- To **find the magnetic field** at a point **outside the capacitor**, let us **apply Ampere's circuital law.**

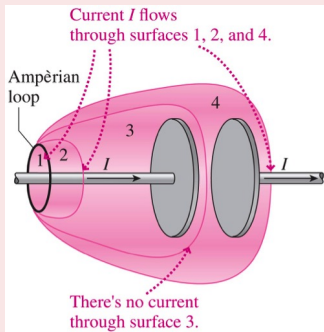
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \dots (1)$$

DISPLACEMENT CURRENT:



- Consider a parallel plate **capacitor C** which is a part of circuit through which a time-dependent **current I** flows.
- To find the **magnetic field at a point P**, in a region outside the parallel plate capacitor, let us **consider a plane circular loop of radius r** whose plane is perpendicular to the direction of the current-carrying wire, and which is **centred symmetrically** with respect to the wire.

DISPLACEMENT CURRENT:

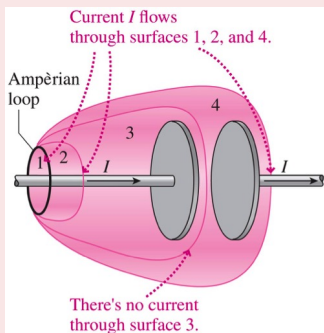


- By symmetry, the **magnetic field is directed along the circumference** of the circular loop and is the **same in magnitude at all points on the loop**. If B is the magnitude of the field, then

$$\oint \vec{B} \cdot d\vec{l} = B \times 2\pi r$$

- There are **infinitely many surfaces that can be attached to any loop**, and **Ampere's law** stated in Equation (1) is **independent of the choice of surface**.

DISPLACEMENT CURRENT:



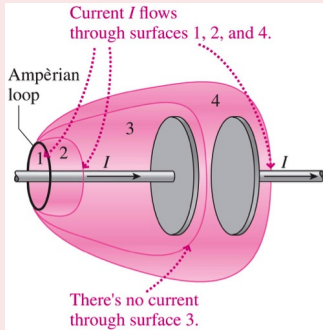
- Surface 1, 2, and 4 gives a **non-zero** value for the enclosed current I , whereas surface 4 gives **zero** for the enclosed current because no current passes through it:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad (\text{For surfaces 1, 2, and 4})$$

$$\oint \vec{B} \cdot d\vec{l} = 0 \quad (\text{For surface 3})$$

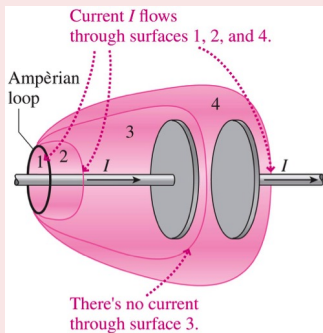
- On applying Ampere's circuital law to such surfaces with the same perimeter, we find that the **LHS term** in eq (2) is the same, but the **RHS term** is **zero**, since no current passes through the surfaces.

DISPLACEMENT CURRENT:



- Calculating by one way, there is a magnetic field at point P. Calculating by another way, the magnetic field at point P is zero. Hence, Ampere's circuital law must be missing something.
- The missing term is due to the passage of electric field through the **surface 4** between the plates of the capacitor.
- If $A \rightarrow$ area of the plates of the capacitor and $Q \rightarrow$ total charge on each plate of the capacitor

DISPLACEMENT CURRENT:



- The electric field E between the plates is given by

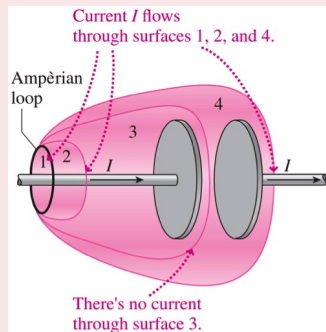
$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{A\epsilon_0} \dots (3)$$

- This **field is perpendicular to the surface** in figure (1). the capacitor plates. Using Gauss's law, the **electric flux through the surface 3** is given by

$$\Phi_E = EA = \frac{Q}{\epsilon_0}$$

Or $Q = \epsilon_0 \Phi_E \dots (4)$

DISPLACEMENT CURRENT:



- Since the **charge Q on the capacitor plates changes with time**, there is a current I which is given by

$$I = \frac{dQ}{dt} = \epsilon_0 \frac{d\Phi_E}{dt} \dots (5)$$

- **This is the missing term in Ampere's circuital law.**
- The current given by eq (5) is due to **changing electric field** (or electric displacement). Hence, it is called '**Displacement Current**'.
- The **current carried by conductors due to flow of charges** is called '**conduction current**'.

Ampere-Maxwell law:

- The generalised (and correct) Ampere's circuital law: "the total current passing through any surface of which the closed loop is the perimeter" is the **sum of the conduction current and the displacement current**.
- The generalised law is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d) = \mu_0 \left(I_c + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

MAXWELL'S EQUATIONS IN VACUUM

1. $\oint \mathbf{E} \cdot d\mathbf{A} = Q / \epsilon_0$ (Gauss's Law for electricity)
2. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$ (Gauss's Law for magnetism)
3. $\oint \mathbf{E} \cdot d\mathbf{l} = \frac{-d\Phi_B}{dt}$ (Faraday's Law)
4. $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 i_c + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$ (Ampere – Maxwell Law)

Exercise 1

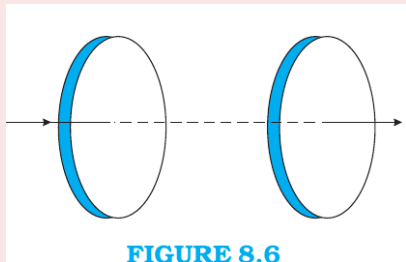


Figure 8.6 shows a capacitor made of two circular plates each of **radius 12 cm**, and **separated by 5.0 cm**. The capacitor is being charged by an external source (not shown in the figure). The charging **current is constant** and equal to **0.15 A**.

- Calculate the capacitance and the rate of change of potential difference** between the plates.
- Obtain the **displacement current** across the plates.
- Is Kirchhoff's first rule (junction rule) valid** at each plate of the capacitor? Explain.

Exercise 2

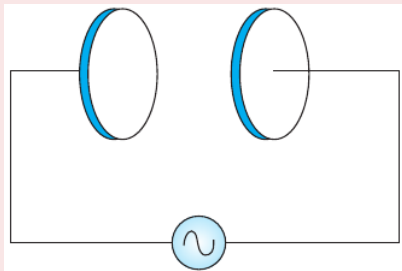


FIGURE 8.7

A parallel plate capacitor (Fig. 8.7) made of circular plates each of **radius $R = 6.0$ cm** has a **capacitance $C = 100$ pF**. The capacitor is connected to a **230 V ac** supply with a **(angular) frequency** of 300 rad s^{-1} .

- (a) What is the **rms value** of the **conduction current**?
- (b) Is the **conduction current** equal to the **displacement current**?
- (c) Determine the **amplitude of B** at a point **3.0 cm from the axis** between the plates.

ELECTROMAGNETIC WAVES

Sources of electromagnetic waves

- It is an important result of Maxwell's theory that **accelerated charges radiate electromagnetic waves**.
- Consider a charge oscillating with some frequency.
- This **produces an oscillating electric field** in space, **which produces an oscillating magnetic field**, which in turn, is a source of oscillating electric field, and so on.
- The **frequency of the electromagnetic wave** naturally **equals the frequency of oscillation of the charge**.
- The **energy associated with the propagating wave comes** at the expense of the **energy of the source** – the accelerated charge.

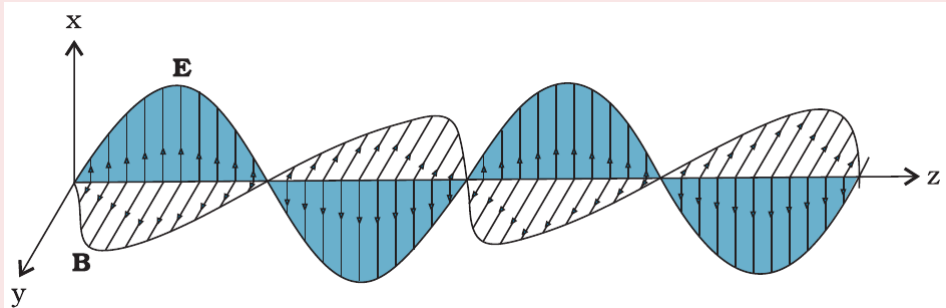
ELECTROMAGNETIC WAVES

Sources of electromagnetic waves

- **Hertz's** successful experimental test of Maxwell's theory
- Seven years after Hertz, **Jagdish Chandra Bose**, working at Calcutta (now Kolkata), succeeded in producing and observing electromagnetic waves of much **shorter wavelength (25 mm to 5 mm)**.
- **Guglielmo Marconi** in Italy followed Hertz's work and **succeeded in transmitting electromagnetic waves** over distances of many **kilometres**.

Nature of electromagnetic waves

- It can be shown from Maxwell's equations that **electric and magnetic fields** in an electromagnetic wave are **perpendicular to each other**, and to the **direction of propagation**.



Nature of electromagnetic waves

- a typical example of a plane **electromagnetic wave propagating along the z direction**
- The **electric field E is along the x-axis**, E_x and varies sinusoidally with z , at a given time.
- The **magnetic field B_y is along the y-axis**, and again varies sinusoidally with z .
- The electric and magnetic fields E_x and B_y are **perpendicular to each other, and to the direction z of propagation**.
- We can write E_x and B_y as follows:

$$E_x = E_0 \sin(kz - \omega t) \quad \text{and} \quad B_y = B_0 \sin(kz - \omega t)$$

Nature of electromagnetic waves

- $E_x = E_0 \sin(kz - \omega t)$ and $B_y = B_0 \sin(kz - \omega t)$
- Here, $k = \frac{2\pi}{\lambda}$ is called **propagation or wave vector**.
- $\omega = 2\pi\nu = \frac{2\pi}{T}$ is **angular frequency**
- **speed of wave** $v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = \lambda\nu$
- Also, $\frac{\omega}{k} = \frac{\frac{2\pi}{T}}{\frac{2\pi}{\lambda}} = \frac{\lambda}{T} = v$

Nature of electromagnetic waves

- Also, speed of EM wave $v = \frac{1}{\sqrt{\mu\epsilon}}$. In Vacuum $c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$
- from Maxwell's equations that the magnitude of the electric and the magnetic fields in an electromagnetic wave are related as

$$E_0 = cB_0$$

- EM waves are **Transverse** and are self-sustaining oscillations of electric and magnetic fields in free space, or vacuum. They differ from all the other waves we have studied so far, in respect that **no material medium is involved in the vibrations of the electric and magnetic fields.**

Example 1

A plane electromagnetic wave of **frequency 25 MHz travels** in free space along the **x-direction**. At a particular point in space and time, $E = 6.3 \hat{j}$ V/m. **What is B at this point?**

Example 2

The **magnetic field** in a plane electromagnetic wave is given by $B_y = (2 \times 10^{-7}) T \sin (0.5 \times 10^3 x + 1.5 \times 10^{11} t)$.

- (a) What is the **wavelength and frequency** of the wave?
- (b) Write an **expression for the electric field**.

Example 2

Suppose that the electric field amplitude of an electromagnetic wave is $E_0 = 120 \text{ N/C}$ and that its frequency is $\nu = 50.0 \text{ MHz}$.

- (a) Determine, B_0 , ω , k , and λ .
- (b) Find **expressions for E and B**.

Electromagnetic Spectrum

- Classification of EM waves according to frequency/wavelength.
- All EM waves travel with the same speed in vacuum:

$$c = 3 \times 10^8 \text{ m/s}$$

- They differ in wavelength, frequency, production, detection, and interaction with matter.
- Order of spectrum (increasing frequency): Radio $\rightarrow$ Microwave $\rightarrow$ Infrared $\rightarrow$ Visible $\rightarrow$ Ultraviolet $\rightarrow$ X-ray $\rightarrow$ Gamma.

Radio Waves

- Wavelength: $> 0.1 \text{ m}$
- Production: accelerated electrons in aerials.
- Detection: receiver antennas.
- Uses: Radio and TV communication, AM/FM, mobile phones (UHF band).

Microwaves

- Wavelength: 0.1 m – 1 mm (GHz range).
- Production: klystron, magnetron, Gunn diode.
- Detection: point-contact diodes.
- Uses: RADAR (navigation, speed guns), microwave ovens (resonant heating of water molecules).

Infrared (IR) Waves

- Wavelength: 1 mm – 700 nm.
- Production: hot bodies, molecular vibrations.
- Detection: thermopiles, bolometer, IR photographic film.
- Uses: heat therapy lamps, remote controls, weather satellites, greenhouse effect.

Visible Light

- Wavelength: 700 – 400 nm, frequency: $4 \times 10^{14} - 7 \times 10^{14}$ Hz.
- Production: electronic transitions in atoms.
- Detection: human eye, photocells, photographic film.
- Uses: vision, information about surroundings.

Ultraviolet (UV) Rays

- Wavelength: 400 nm – 1 nm.
- Production: Sun, hot bodies, special lamps.
- Detection: photocells, photographic film.
- Uses: sterilization, LASIK eye surgery, tanning, water purifiers.
- Note: mostly absorbed by ozone layer; excess exposure harmful.

X-rays

- Wavelength: $1 \text{ nm} - 10^{-3} \text{ nm}$.
- Production: bombard metal target with high-energy electrons (X-ray tube).
- Detection: photographic film, Geiger tube, ionisation chamber.
- Uses: medical imaging, cancer treatment, checking flaws in metals.
- Caution: harmful if overexposed.

Gamma Rays

- Wavelength: $< 10^{-3}$ nm (highest frequency).
- Production: nuclear reactions, radioactive decay.
- Detection: same as X-rays.
- Uses: cancer treatment, sterilization.
- Hazard: highly penetrating, dangerous to cells.

Quick Revision Table

Type	Wavelength	Production	Uses
Radio	$> 0.1 \text{ m}$	Accelerated charges in aerials	Radio, TV, mobile communication
Microwave	$0.1 \text{ m} - 1 \text{ mm}$	Klystron, magnetron, Gunn diode	Radar, ovens, satellite communication
Infrared	$1 \text{ mm} - 700 \text{ nm}$	Hot bodies, molecular vibrations	Heat therapy, remotes, weather satellites, greenhouse effect
Visible	$700 - 400 \text{ nm}$	Atomic electronic transitions	Human vision, photographic film, photocells
Ultraviolet	$400 - 1 \text{ nm}$	Sun, special lamps, hot bodies	Sterilization, LASIK surgery, tanning, water purifiers
X-rays	$1 - 10^{-3} \text{ nm}$	Bombard metal target with high-energy electrons	Medical imaging, cancer treatment, metal flaw detection
Gamma	$< 10^{-3} \text{ nm}$	Nuclear reactions, radioactive decay	Cancer therapy, sterilization (high penetration)

Mnemonic for Spectrum

- Easy way to remember order:
- **“Ravi’s Mother Is Visiting Uncle Xavier’s Garden.”**
- Mapping:
 - Ravi’s → Radio
 - Mother → Microwave
 - Is → Infrared
 - Visiting → Visible
 - Uncle → Ultraviolet
 - Xavier’s → X-rays
 - Garden → Gamma